

OMR Examination System Workflow

1. Project Goal

Create a student registration and OMR evaluation system for a general knowledge test scheduled for 10 August 2026. The system

- student registration using registration number or form number
- automatic student data lookup
- manual fallback entry for students without prior results
- generation of a unique 7-digit examination number
- OMR metadata preparation for each submitted student
- invigilator upload of scanned OMR sheets to a folder
- automatic OCR/OMR processing with confidence-based validation
- admin review for flagged responses
- final results pushed to Firestore for admin dashboard visibility

2. Core Roles

Student

- enters registration number or form number
- fetches latest registration details
- confirms submission
- receives exam number

Invigilator

- logs in with email or mobile number
- uploads scanned OMR images
- monitors upload status

Super Admin

- manages system configuration
- reviews flagged sheets
- approves final results
- downloads/prints prepared OMR templates

3. Registration Workflow

1. Student opens the registration page.
2. Student enters registration number or form number.
3. System looks for the latest student record.
4. If a match is found, the system pre-fills:
 - form number
 - session
 - registration number
 - class roll number
 - latest class
 - photo
 - name
 - father name
5. Student confirms the details.
6. System creates a unique 7-digit examination number.
7. The submission is stored in Firestore.

Manual fallback

If no result is found:

- student can enter details manually
- required fields:
 - name
 - father name
 - class
 - class roll number
- session defaults to 2025-26
- photo upload is disabled
- exam number is still generated

4. OMR Preparation Workflow

For each valid submission:

- backend prepares an OMR record containing: